

Introduction to CV

Part II

Today's Agenda

Gamma Correction

Image Filtering

Convolution

Gaussian Blur

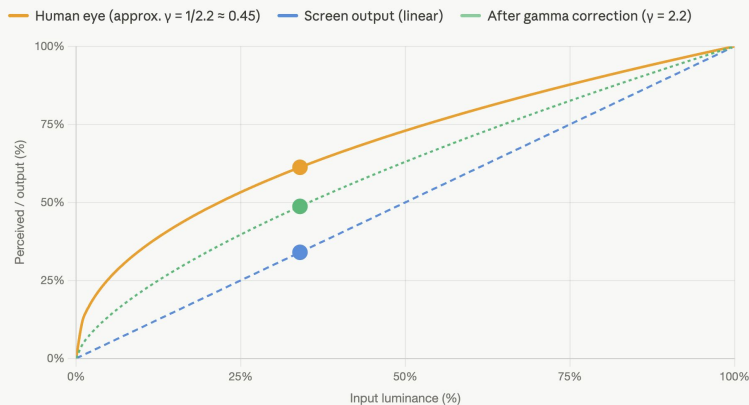
Median Blur

Q&A and Coding

Gamma Correction

What is gamma correction?

- Gamma correction is a nonlinear adjustment applied to images to control the brightness and contrast perception.
- Human vision perceives brightness **logarithmically**, while screens display it **linearly**.
- Without correction, images may appear too dark or too bright.



The Gamma Correction Formula

The gamma correction operation is applied pixel-wise using the formula.

$$p' = f(p) \longrightarrow \text{red square} = f(\text{brown square})$$

That means that the operation is applied to each pixel of the image separately

The Gamma Correction Formula

$$I_{out} = I_{in}^{\gamma}$$

Where:

- I_{in} = Input pixel intensity (normalized between 0 and 1)
- I_{out} = Corrected pixel intensity
- γ = Gamma value (typically **2.2 for monitors**)

The Gamma Correction Formula

If $\gamma > 1$, the image becomes darker.

If $\gamma < 1$, the image becomes brighter.



Linear RAW Image
(image gamma = 1.0)



Gamma Encoded Image
(image gamma = 1/2.2)

Image Filtering

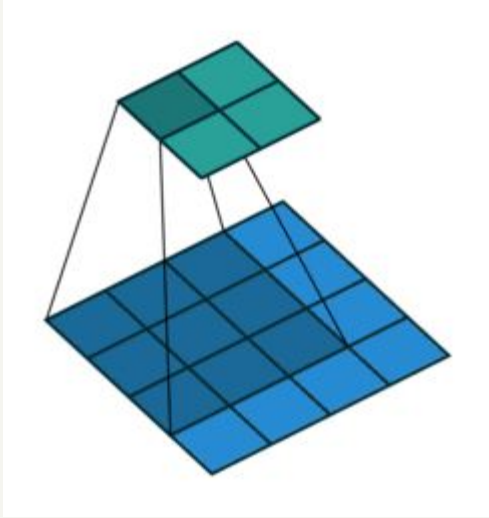
Image Filtering

$$p' = f(p) \longrightarrow \text{red square} = f\left(\begin{array}{|c|c|c|} \hline \text{yellow} & \text{brown} & \text{brown} \\ \hline \text{brown} & \text{brown} & \text{orange} \\ \hline \text{orange} & \text{brown} & \text{orange} \\ \hline \end{array} \right)$$

These operations are used for

- Improving image quality
- Image manipulation
- Feature detection

Convolution



- Convolution is a key image processing technique where a **kernel (filter)** slides over an image, modifying each pixel based on its neighbors.
- It is used for **blurring, sharpening, and edge detection** by computing a weighted sum of pixel values.

Convolution

0	2	1	0	1
0	1	2	1	0
1	0	2	0	0
1	0	0	1	1
0	1	1	2	2

Input Vector

0	1	-4
0	2	0
-1	4	3

Filter

5		

Output

$$\begin{aligned} &= 0 \times 0 + 2 \times 1 + 1 \times -4 + \\ &\quad 0 \times 0 + 1 \times 2 + 2 \times 0 + \\ &\quad 1 \times -1 + 0 \times 4 + 2 \times 3 = 5 \end{aligned}$$

High Pass Filter (sharpening)



$$* \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix} =$$



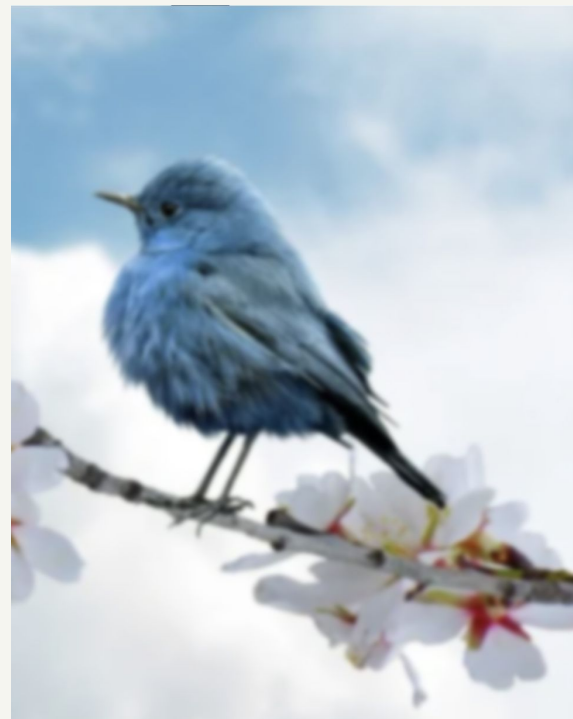
`= original + 4 × (original - average of neighbours)`

Lower Pass Filter (blurring)



$$* \frac{1}{6} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix} =$$

Low-pass filter
(blurring)



Gaussian Blur

Gaussian Blur

- Gaussian Blur is a smoothing technique used in image processing to reduce noise and detail.
- A **Gaussian kernel** (matrix) is applied to each pixel.
- Neighboring pixel values are weighted based on a **Gaussian distribution**.
- The result is a blurred image with reduced sharp edges and noise.

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

where σ (**sigma**) controls the blur intensity. A higher σ results in stronger blurring.

Gaussian Blur

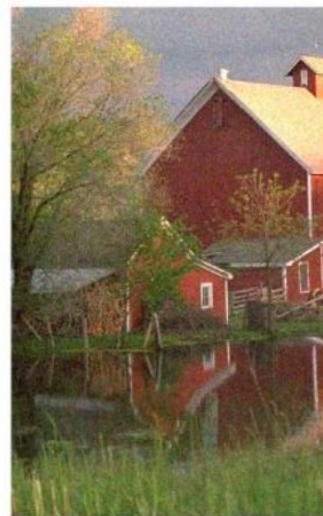
Original



Noisy



Gaussian filtering



$\sigma = 0.1$



$\sigma = 1$



$\sigma = 3$

Median Blur

Salt and Pepper Noise



- Gaussian Blur does not smooth out nicely the salt and pepper noise.
- In this case we need to use another approach - **median blur**

Median Blur

- Unlike Gaussian Blur, which averages pixel values, Median Blur **replaces each pixel with the median value** from its surrounding neighbors.
- A **kernel (window)** moves across the image.
- All pixel values inside the kernel are **sorted**.
- The **middle (median) value** replaces the central pixel.

<Coding time>

Q&A time!